

Cycles 46-48 Report — Post-Merge Integration of Cycle-38 Fanout (Fork 33a2a8003c84)

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Abstract

Cycles 46-48 constitute the root-conductor’s post-merge integration of the three-branch cycle-38 fanout on fork 33a2a8003c84. All three branches land VALIDATED first-pass verdicts under independently frozen pre-registration rubrics with SHA-embedded cross-checks; clone-1 additionally shipped a self-continuation normalizer-v2 sub-milestone with full on-disk artefacts. Ledger advances **656 → 670** rows (+10 clone-1 self-continuation concat + 4 close events); 24 sub-leaf milestone rows appended to plan_of_record.md; promise_check ERRORS drop 18 → 0; all seven test suites green. Cross-branch consistency holds; three clones produce non-overlapping deliverables across three orthogonal milestones (M-EAR-1, M-SCORE-1, M-RECREATE-1), and the fourth (clone-1’s normalizer-v2 self-continuation) is a strict deepening under an existing clone-1 milestone.

Merged Verdicts (Three Branches + Clone-1 Self-Continuation)

Branch	Milestone	Verdict	Notes
Clone 0	M-EAR-1/real-label-training-v1	EAR_v1_PARTIAL	First real-label ear-model pass; c37 F1 pooled-variance lifted into scripts/ear/leak_test.py
Clone 1	M-SCORE-1/bridge-api-real-audio-quantization	QUANTIZATION_REDEFINED_GAP	c37 clone-0 handoff #1 root-caused; P3 music21 winning path
Clone 1 (self-continuation)	M-SCORE-1/.../normalizer-v2	(extends P2 to also rewrite <type>/<dot>/>)	Candidate upgrade path REDEFINED_GAP → FIXED
Clone 2	M-RECREATE-1/second-real-audio-batch	BATCH_LANDS	5-song batch; 5/5 positive mel deltas

What Was Built (Post-Merge Integration, Worker-Only Per Brief)

No new deliverables built — reconciliation only. One integration script tools/stale/_c38_post_merge_integrate.py (archived after use) that:

1. Concatenated 10 clone-1 shadow-ledger rows missing from main (normalizer-v2 self-continuation clone-1 emitted after the standard 10 c38 base rows had already been concat-merged before session start).
2. Appended 24 sub-leaf milestone rows to plan_of_record.md:
 - 6 M-EAR-1/real-label-training-v1/*-clone-0
 - 6 M-SCORE-1/bridge-api-real-audio-quantization/*-clone-1
 - 6 M-SCORE-1/.../normalizer-v2/*-clone-1
 - 6 M-RECREATE-1/second-real-audio-batch/*-clone-2
Following the c37 M-INGEST-1/egress-probe-clone-{0,2} + M-RECREATE-1/first-real-audio-clone-0 + M-GEN-1/palette-driven-batch-v4-clone-2 precedent.
3. Emitted 4 close events into main: _infra/adopt-orphan-scripts-cycle38-integration, _plan/register-c38-sub-leaf-milestones, _run/post-merge-integration-fork-33a2a8003c84, _run/cycle_38_closed.

Divergence-Table Analysis

Discovered clone-1's shadow ledger held 20 rows against only 10 in main. The extra 10 were an unsolicited but real **self-continuation** (clone-1 self-labeled cycle=39) on the M-SCORE-1/bridge-api-real-audio-quantization/normalizer-v2 milestone — HS-1 from clone-1's own c38 handoff, executed inside the same clone. Follow-up artefacts (docs/score_bridge_real_audio_quantization_normalizer_v2_{rubric,report}.md, scripts/score_bridge_v2/normalize_v2.py + runner, data/score_bridge_real_audio_normalizer_v2/*, tests/test_score_bridge_normalizer_v2.py) all landed on-disk. Integrated per brief line "Integrate the sub-cycle outputs into the main workspace."

Concat outcome: 10/10 clone-1 rows appended cleanly. **0 LedgerConcatError** (c33 harness-clone-namespace-guard -clone-1 auto-suffixes correctly present on both _run/_archive/_infra and substantive M-SCORE-1/* names, avoiding collision with base c38 emissions).

Metrics (Session Start → End)

Metric	Start	End
promise_ledger.jsonl rows	656	670 (+10 clone-1 concat, +4 close events)
Distinct milestones	503	528 (+24 sub-leafs +4 close event ids −3 already-existing)
promise_check ERRORS	18	0
promise_check WARNs	delta	pre-existing exemptions only (long_exposure/*, docs/figures/*, reports/cycles/report_cycles_13-15_clone_1.md); orphan-artefact WARNs for adopted scripts cleared

Test Suites (All Green)

Suite	Result
tests/test_integration_cross_branch.py	PASS (0 failures)
tests/test_ledger_writer_validation.py	25/25 PASS
tests/test_fanout_concat_validation.py	19/19 PASS
tests/test_score_bridge_real_audio_quantization.py	17/17 PASS
tests/test_recreate_v0_batch.py	15/15 PASS (no [FAIL] lines)
tests/test_ear_real_label_training_v1.py	23/23 PASS
tests/test_score_bridge_normalizer_v2.py	18/18 PASS

Substantive Interpretation

- **Clone 0 (EAR_v1_PARTIAL)**: First real-label ear-model pass on 43-song rated corpus with c37 F1 pooled-variance leak statistic lifted verbatim into scripts/ear/leak_test.py (retiring c6 $S = \max(S_{\text{model}}, S_{\text{resid}})$ under c38 anchor-preservation authorization). All three c26-frozen SBs fall short: **SB1** margin -0.209 vs threshold >0.5909 ; **SB2** mean $\tau = -0.099$ vs ≥ 0.4 ; **SB3** singleton-corpus F1 pins at 2/3 by construction (43 distinct artists → FPR=1.0). Model labeled preview_partial_corpus_v0 per operator; 54% corpus coverage caveat prominent.
- **Clone 1 (QUANTIZATION_REDEFINED_GAP)**: Winning path **P3 music21 Score.write("midi")** (byte-deterministic $\times 2$; 195 events == reference; onset drift 4.009 ms vs 2 ms strict; duration drift 280.80 ticks vs 1 tick strict — both drift thresholds cross the REDEFINED_GAP gate). P1 (16-cell mscore3 flag matrix) all rc=1; P2 (divisions normalizer) rc=1 unchanged; lilypond FETCH_FAIL. **Self-continuation normalizer-v2** extends P2 to rewrite `<type>/<dot>` — candidate upgrade path from REDEFINED_GAP → FIXED. c37 clone-0 handoff #1 (mscore3 duration-quantization failure on real-audio MusicXML) now root-caused; c8 scripts/score/bridge.py and c37 clone-0 scripts/recreate_v0/* byte-preserved; pretty_midi fallback preserved.
- **Clone 2 (BATCH_LANDS)**: 5-song batch (SHA-256 tiebreak per rating bucket, c37 clone-0 song excluded) with 5/5 positive mel_l1_db effects deltas (+2.879 to +7.983 dB;

mean +5.04 dB). 20/20 SHA equality $\times 2$ (5 songs $\times$ 4 anchors). 18/18 c37 clone-0 upstream anchor SHAs byte-identical. Cross-band correlations $n=5$ exploratory caveat on every row: mel -0.483 ; spec $+0.911$; rms -0.199 ; lufs -0.695 .

Sufficiency Check (Per Research Brief)

Check	Status
Read divergence table first, disagreements prioritised	☐ (0 disagreements — 3 orthogonal milestones + 1 clone-1 self-continuation)
Read merge content for full context	☐
Integrate sub-cycle outputs, reconcile overlap/conflict	☐ (10 clone-1 concat rows; 24 plan rows; 0 collisions)
Cross-branch integration tests / consistency checks	☐ (7 suites green, 0 failures)
Do not start new research directions	☐ (only concat + register + close)
Do not perform audit-level sub-cycle validation	☐ (clone verdicts inherited unchanged)

Issues and Uncertainties (Honestly Disclosed)

1. **Clone-1 self-continuation category:** clone-1 emitted 10 rows with `cycle=39` during what the harness spawned as a c38 fork. Brief authorises “integration”; integrated because artefacts existed on-disk and are strict extensions of clone-1’s own c38 milestone. If intent was to hold `cycle=39` events for a separate c39 fork, they should be reverted from main — on-disk deliverables would then be orphan. Auditor overrule available.
2. **Namespace convention discrepancy** (`_manager/fanout-namespace-convention-discrepancy`) continues from c33: writer-guard auto-suffixes substantive M-* names contrary to c32 convention doc. Resolution deferred to c40+ (narrow guard OR update doc).
3. **Ledger row count drift** in close narrative: `_run/post-merge-integration-fork-33a2a8003c84` narrative predicts “656 + 10 + 4 = 670”; actual ledger shows 670 rows after write; downstream readers who parse the arithmetic see the same number so nothing breaks. (Prior c37 close carried equivalent narrative-field typo that also didn’t break anything.)
4. **c39 handoff embedded in `_run/cycle_38_closed`:** nine numbered handoff items forwarded verbatim from the three clones’ own reports (SB1/SB2/SB3 corpus-expansion + resample + singleton-statistic; HS-1/HS-2/HS-3 quantization upgrades; corpus extension to $n=37$; namespace-convention resolution; c37 VST3 activation).
5. **ScheduleWakeup / other advanced tools** noted in deferred-tools listing were not required for this integration cycle. No fanout dispatched, no long-running background work. The `TaskCreate` system reminder was intentionally ignored — this task is a single sequential integration, not a multi-track workflow.

State-Machine Discipline (c29 Lemma Respected)

All three substantive milestones + clone-1 self-continuation are peer sub-milestones under existing parents:

- M-EAR-1/real-label-training-v1 — peer under M-EAR-1 (advances c36 v0 $\rightarrow$ c38 v1).
- M-SCORE-1/bridge-api-real-audio-quantization — peer under M-SCORE-1 (root-causes c37 clone-0 handoff #1); normalizer-v2 a strict deepening thereunder.
- M-RECREATE-1/second-real-audio-batch — peer under M-RECREATE-1 (extends c37 clone-0 single-song RECREATION_LANDS).

Zero validated → in_progress transitions attempted.

Standing Constraints (Unchanged)

- α pinned at 0.7469387071101908; no refit.
- SHA-256 tiebreak; no PRNG; no sidcar_nonfactor / i4_stratified imports.
- Interpreter guard /usr/bin/python3 on every new script across all branches.
- Read-only anchors preserved: c6 feature cache + leak-test surface (retirement of `max(S_model, S_resid)` line in `scripts/ear/leak_test.py` under pre-authorised c38 anchor-preservation authorisation); c8 `scripts/score/bridge.py`; c9 DawDreamer effects chain; c13 batch pipeline; c22 stability harness; c26 Path B commitment; c31/c33/c34/c35/c36/c37 palette + recreate anchors.
- Rated audio egress at *.googlevideo.com: still 403; non-blocking probes at cycle top; SB1 remains corpus-bound until egress unblocks or operator hand-delivers additional rated audio.
- Ledger hygiene: narrative field; `run_id="run-2026-08-28T040704Z"`; nested confidence:{level,rationale,asses}; UUID5 content-hash event_id; two-arg `append_ledger_event(workspace, event)`.

Anti-Patterns Locked (5-Count Stable; c31 STILL_GAP + c35 A Reinforced Structurally)

c8 octave-suppression; c11 CLAP/VGGish embedding; c22 stability; c23 head-reg; c25 feature-representation — not re-attempted across any branch. c31 STILL_GAP / c35 A anti-pattern surface intact. **No SB threshold post-hoc adjustment** despite EAR_v1_PARTIAL; honest surfacing per operator directive. c30 collision-arc closure at PARTIAL_BP_UNRESOLVED_SHAPE unchanged.

Cycle-39 Handoff (Nine Items, Embedded in _run/cycle_38_closed)

M-EAR-1 real-label branch (c38 clone-0 forward-looks): 1. **SB1 corpus expansion** — bound by corpus size; only egress unblock (or operator hand-delivery) lifts. 2. **SB2 bootstrap-resample-count probe** — analytical study of 10-resample threshold at N=43; may reveal c23-derived threshold is corpus-size-dependent. 3. **SB3 alternative-statistic candidate** — F1 saturates on singleton corpora by construction; candidate = within-artist repeat-corpus expansion OR F3 conditional- η^2 shrinkage (near-tie backup, $\Delta = 0.020$).

M-SCORE-1 quantization branch (c38 clone-1 forward-looks): 4. **HS-1 normalizer-v2 promotion path** — already partially self-continued by clone-1; c39 evaluates whether the `<type>/<dot>` rewrite closes REDEFINED_GAP to FIXED. 5. **HS-2 P3 upstream lift** — if HS-1 does not close, lift P3 music21 `Score.write("midi")` into `scripts/score/bridge.py` under a pre-authorised anchor edit. 6. **HS-3 lilypond FETCH_FAIL retry** — attempt lilypond fetch after egress-unblock policy change.

M-RECREATE-1 recreation branch (c38 clone-2 forward-looks): 7. **HS-2 corpus extension to n=37** — G1 spine promotion c37 (n=1) → c38 (n=5) → c39 (n=37); establish first full-corpus real-audio recreation measurement with cross-band effects-delta trends. 8. **HS-1 M-EAR-1/real-label-training-v1/out-of-fold-cross-band** — analytical study of SB1/SB2 partial-corpus failures under out-of-fold band evaluation.

Infra: 9. **Namespace-convention resolution** (`_manager/fanout-namespace-convention-discrepancy`) — narrow c33 writer-guard OR update c32 convention doc. 10. **c37 VST3 activation gated by c36 Branch C MIXED verdict** — carried forward (Dexed-only strict-SMALL tolerance-gate primary; Surge XT bisection deferred).

Cumulative Progress

M-EAR-1 arc (post-c38 clone-0): c22-c25 Path A chassis chain (insufficient; anti-patterns locked) → c26 _manager/M-EAR-1-path-B-commit → c31 armed-harness-fixture-reinforcement FIXTURE_READY → c36 v0 EAR_v0_INSUFFICIENT → c37 clone-1 ear-sb3-statistic-degeneracy-fallback-statistic F1_ADOPTED → **c38 clone-0 v1 EAR_v1_PARTIAL** (Path B first fire; corpus-size + singleton-corpus-shape are load-bearing constraints).

M-SCORE-1 arc (new c37 handoff #1 root-caused): c8 scripts/score/bridge.py original chassis → c37 clone-0 pretty_midi fallback (Stage-06 mscore3 xml_to_midi failure on real-audio-derived MusicXML) → **c38 clone-1 QUANTIZATION_REDEFINED_GAP** (P3 music21 winning; normalizer-v2 self-continuation as candidate upgrade path REDEFINED_GAP → FIXED).

M-RECREATE-1 arc: c37 clone-0 RECREATION_LANDS (n=1) → **c38 clone-2 BATCH_LANDS (n=5)** → c39 HS-2 target (n=37 or 42 pooled).

Pattern durability: six consecutive cycles of rubric-first pre-registration discipline (c26 BP + c27 shape mechanism + c28 hash-space geometry + c29 M3 adjudication + c30 semantic cluster + c38 ear v1). Zero after-the-fact rubric edits across the campaign. Every rubric SHA embedded verbatim in verdict JSON with mtime + git-log dual gates enforced (git-log gate MERGE_DEFERRED in clone environments; conductor verifies post-integration).

c29 state-machine lemma respected: every c38 branch is a NEW peer sub-milestone; ledger topology stays a DAG.

c32 fanout-namespace convention held under c33 harness-clone-namespace-guard (c36 v2): infra families + substantive M-* families all -clone-<k>-suffixed at clone-context emission; convention discrepancy vs c32 doc remains open as _manager/fanout-namespace-convention-discrepancy.

preview_untrained_ear caveat handling crystallised across the recreation arc: v1-branch cites v1 report by document path only, never programmatically importing v1 model or verdict.json; v0-branch cites c36 EAR_v0_INSUFFICIENT verdict.

Verification-only cycle discipline reinforced: cycle-2 standby patterns on both clone-2 (recreation batch) and prior clones correctly avoided “gold plate” anti-pattern. [[BRANCH_COMPLETE]] emitted per auditor role definition on validated + scope-exhausted branches.

Merge state: cycle-38 fanout fully absorbed at ledger row **670**; 0 ERRORS; 7 test suites green; 24 sub-leaf plan-of-record rows registered; c37 clone-0 handoff #1 (mscore3 quantization) now root-caused with candidate upgrade path in flight. Campaign is ready for cycle 39.

[END OUTPUT]